x11vnc Service - Installation Guide

Share the currently logged-in X11 desktop on a Dell PC running Debian-based Raspberry Pi Desktop.

Overview

The install-x11vnc.sh script installs x11vnc and xauth, configures a VNC password, and creates a systemd service named x11vnc-follow-session.service. The service starts at boot and waits for the active local X11 user session.

Unlike a virtual VNC desktop, this displays the same physical desktop that the logged-in user sees.

Requirements

Debian-based Raspberry Pi Desktop on a Dell/PC; an X11 graphical session; sudo/root permissions; network access for installing packages. It does not require Raspberry Pi hardware.

1. Install

Recommended - SSH tunnel mode: Only listens on 127.0.0.1 by default.

```
sudo bash install-x11vnc.sh
```

Optional - direct local network mode: Listens on TCP port 5900 on network interfaces.

```
sudo bash install-x11vnc.sh --lan
```

On the first run, you will be asked to set a VNC password. Traditional VNC authentication uses only the first 8 characters. Re-running the installer keeps the existing password unless you reset it.

2. Connect to the desktop

Default secure mode: An SSH server must also be running on the Dell. On the remote computer, open a tunnel, then point your VNC viewer at 127.0.0.1:5900.

```
ssh -L 5900:127.0.0.1:5900 USER@DELL_IP
```

LAN mode: Find the Dell IP with hostname -I, then connect the viewer to DELL_IP:5900.

> Important: A user must be logged in to an active X11 desktop. The service will wait when no qualifying X11 desktop is active; it does not share the login screen.

Operations & Troubleshooting
Useful commands and behavior for the installed systemd service.

3. Manage the service

```
sudo systemctl status x11vnc-follow-session
sudo systemctl restart x11vnc-follow-session
sudo systemctl stop x11vnc-follow-session
sudo systemctl enable --now x11vnc-follow-session
```

Follow live diagnostic logs:

```
sudo journalctl -u x11vnc-follow-session -f
```

4. How active-session detection works

The helper checks loginctl about every 3 seconds and selects a local, active, user-class X11 session on seat0. It identifies the X display and Xauthority credentials, starts x11vnc against that display, and restarts x11vnc when the qualifying session changes. If no session qualifies, it waits.

5. Files installed

/etc/systemd/system/x11vnc-follow-session.service	Systemd unit (starts at boot).
/usr/local/sbin/x11vnc-follow-session	Session-aware launcher.
/etc/default/x11vnc-follow-session	Binding mode: local or lan.
/etc/x11vnc.pass	Password file (root-only permissions).

6. Troubleshooting

VNC not connecting? Check that the GUI uses X11, that somebody is logged in locally, and that the service logs show an active session:

```
echo $XDG_SESSION_TYPE
loginctl list-sessions
sudo journalctl -u x11vnc-follow-session -n 40 --no-pager
```

If the first command reports wayland, choose an X11 session; x11vnc cannot share a native Wayland session. If the service detects a session but cannot display it, inspect the logs for X display or Xauthority errors.

7. Security and configuration

By default, port 5900 is bound to localhost. To change modes, edit /etc/default/x11vnc-follow-session and set X11VNC_BIND=local or X11VNC_BIND=lan, then restart the service.

Security: Direct VNC traffic is not encrypted. Prefer the SSH tunnel; if using LAN mode, restrict port 5900 with a firewall and never expose it directly to the public internet.